

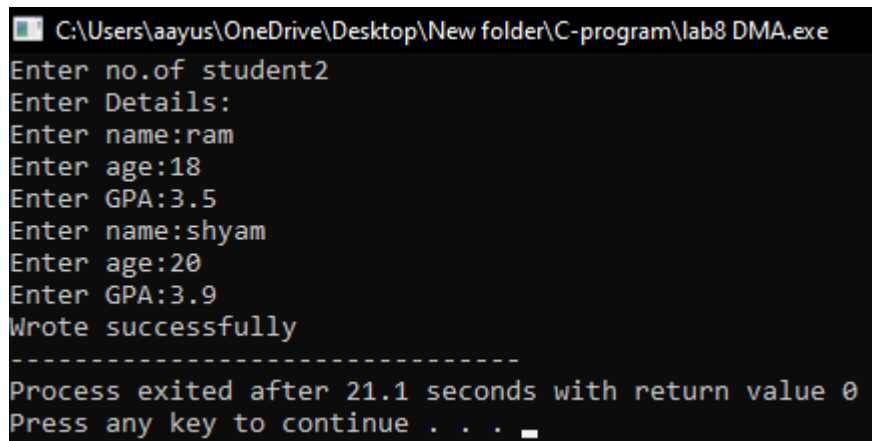
Lab-9 (File Handling)

1. Write a c-program to read name, age and GPA of student from user and store it into file name student.txt.

Ans:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int age;
    char name[50];
    float GPA;
    FILE *p;
    p=fopen("student.txt","w"); //for opening file
    int i,n;
    printf("Enter no.of student");
    scanf("%d",&n);
    printf("Enter Details:\n");
    for(i=0; i<n; i++)
    {
        printf("Enter name:");
        scanf(" %50[^\n]s",name);
        printf("Enter age:");
        scanf("%d",&age);
        printf("Enter GPA:");
        scanf("%f",&GPA);
        fprintf(p,"Name is %s Age is %d GPA is %f",name,age,GPA);
    }
    printf("Wrote successfully");
    fclose(p);
}
```

Output:



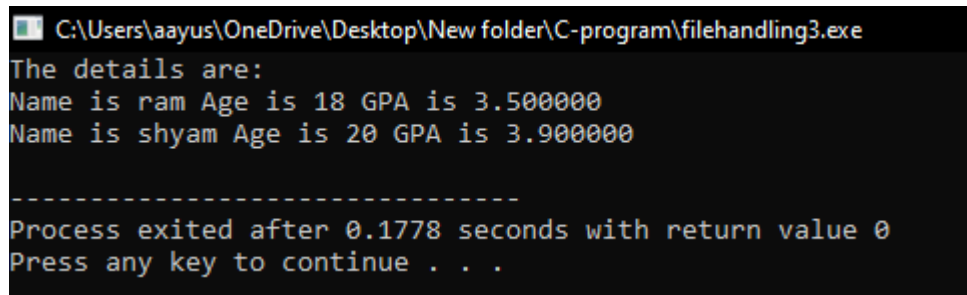
```
C:\Users\ayus\OneDrive\Desktop\New folder\C-program\lab8 DMA.exe
Enter no.of student2
Enter Details:
Enter name:ram
Enter age:18
Enter GPA:3.5
Enter name:shyam
Enter age:20
Enter GPA:3.9
Wrote successfully
-----
Process exited after 21.1 seconds with return value 0
Press any key to continue . . .
```

2. Write a c-program to read that created file above student.txt.

Ans:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int age;
    char name[50];
    float GPA;
    FILE *p;
    p=fopen("student.txt","r");
    if(p==NULL)
    {
        printf("File not found");
        exit(1);
    }
    printf("The details are:\n"); //for opening file
    while(fscanf(p,"Name is %s Age is %d GPA is %f",name,&age,&GPA)==2)
    {
        printf("Name is %s Age is %d GPA is %f\n",name,age,GPA);
    }
    fclose(p);
    return 0;
}
```

Output:



```
C:\Users\ayus\OneDrive\Desktop\New folder\C-program\filehandling3.exe
The details are:
Name is ram Age is 18 GPA is 3.500000
Name is shyam Age is 20 GPA is 3.900000
-----
Process exited after 0.1778 seconds with return value 0
Press any key to continue . . .
```

3. Write a c-program to read name, id and salary of 3 employee from user and store it into file name emp.txt.

Ans:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int id;
    char name[50];
    float salary;
```

```

FILE *p;
p=fopen("emp.txt","w");
printf("Enter Details:\n");
int i;
for(i=0; i<3; i++)
{
printf("Enter name:");
scanf(" %s",name);
printf("Enter id:");
scanf("%d",&id);
printf("Enter salary:");
scanf("%f",&salary);
fprintf(p,"Name:%s ID:%d Salary:%f",name,id,salary);
}
fclose(p);

p=fopen("emp.txt","r");
//for checking file is found or not
if(p==NULL)
{
printf("File is not created");
exit(1);
}
while(fscanf(p,"Name:%s ID:%d Salary:%f",name,&id,&salary)!=EOF)
{
printf("Name:%s ID:%d Salary:%f\n",name,id,salary);
}
fclose(p);
return 0;
}

```

Output:

```

C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\filehandling3.exe
Enter Details:
Enter name:ram
Enter id:101
Enter salary:32005.5
Enter name:shyam
Enter id:102
Enter salary:31000
Enter name:govinda
Enter id:103
Enter salary:20000
Name:ram ID:101 Salary:32005.500000
Name:shyam ID:102 Salary:31000.000000
Name:govinda ID:103 Salary:20000.000000

-----
Process exited after 41.3 seconds with return value 0
Press any key to continue . . .

```

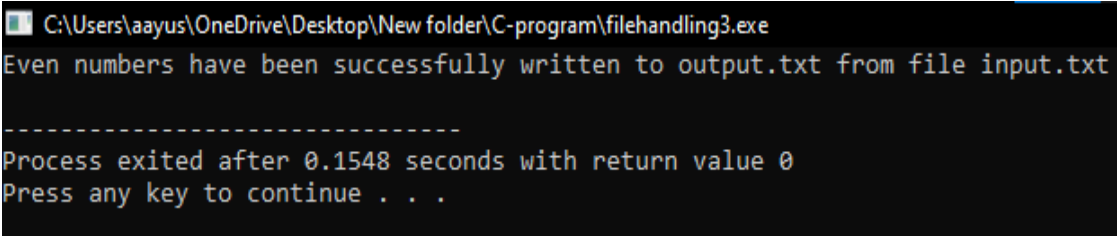
4. Write a c program that reads integer data from file 'input.txt' and write even number from the data to 'output.txt' file.

Ans:

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    FILE *p, *q;
    int num;
    p = fopen("input.txt", "r"); // Open input file for reading
    if (p == NULL) {
        printf("File not found\n");
        exit(1); // Exit if input file doesn't exist
    }

    q = fopen("output.txt", "w"); // Open or create automatically output file for writing
    if (q == NULL) {
        printf("Error opening output file\n");
        fclose(p);
        exit(1);
    }
    // Read integers one by one until end of file
    while (fscanf(p, "%d", &num) != EOF) {
        // Check if number is even
        if (num % 2 == 0) {
            fprintf(q, "%d ", num);
        }
    }
    printf("Even numbers have been successfully written to output.txt from file input.txt\n");
    fclose(p);
    fclose(q);
    return 0;
}
```

Output:



```
C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\filehandling3.exe
Even numbers have been successfully written to output.txt from file input.txt
-----
Process exited after 0.1548 seconds with return value 0
Press any key to continue . . .
```